

Strict p -Dirichlet positivity for Poisson-type quantum generators a partial answer to arXiv:2207.06422

Candidate partial-result packet

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Status: candidate partial result likely valid. The open strict-positivity assertion is proved for primitive Poisson-type generators and positive sums of them. This includes the paper's explicit primitive $[\sigma]_{p,0}$ -DBC example that is not GNS detailed balanced. The general infinitesimal problem remains open.

1 Source question

Li and Lu ask whether, for $1 < p < 2$, every primitive Lindbladian $\mathcal{L}$ satisfying $[\sigma]_{p,0}$ detailed balance obeys

$$\mathcal{E}_{p,\mathcal{L}}(\Gamma_\sigma^{-1}(\rho)) > 0 \quad (\rho > 0, \rho \neq \sigma). \quad (1)$$

They prove this under GNS detailed balance, and for $p = 2$ under the weaker $[\sigma]_{2,0}$ condition, but leave the general case open [1].

the Beckner constant for Markov chains in terms of the Ricci curvature lower bound as in [55].

- For the numerical computation of functional inequality constants, there is a recent research line attempting to estimate the classical log-Sobolev constant by the sum-of-squares relaxation [56]. It is interesting to extend their method to quantum functional inequalities.

We conclude with some interesting and important open questions.

- There are two technical questions we are unable to solve. The first one is stated in Remark 4.7: for a primitive Lindbladian $\mathcal{L}$ satisfying $[\sigma]_{p,0}$ -DBC,

$$\mathcal{E}_{p,\mathcal{L}}(\Gamma_\sigma^{-1}(\rho)) > 0, \quad \text{for } \rho \in \mathcal{D}_+(\mathcal{H}) \text{ with } \rho \neq \sigma. \quad (5.1)$$

The case of $p = 1$ has been proved in [29, Proposition 6.2], from which we easily see that the key step for (5.1) is the convexity of p -Dirichlet form $\mathcal{E}_{p,\mathcal{L}}$ which seems open. Another one is mentioned at the end of Section 4: the monotonicity $\langle \Phi(A), [\Phi(\rho)]_{p,\omega}^{-1} \Phi(A) \rangle \leq \langle A, [\rho]_{p,\omega}^{-1} A \rangle$ under quantum channels with σ -GNS DBC, for which the characterization for such channels [17, Lemma 13] might be helpful. Addressing these questions would directly make our results more complete.

- One important feature of the classical Beckner's inequality is its ability to capture the tail behavior of the family of probability measures: $d\mu_\alpha(x) = c_\alpha e^{-(1+x^2)^{\alpha/2}} dx$ on $\mathbb{R}$ with $\alpha \in [1, 2]$, which satisfies the Poincaré inequality for all α but satisfies LSI only for $\alpha = 2$ [11, 14, 75]. In detail, [75] considers the following variant of (1.7): for $a \in [0, 1]$ and some $C_a > 0$,

$$\sup_{g \in \mathcal{D}_+(\mathcal{H})} \frac{1}{(2-a)^a} (\mu[g^2] - \mu[g^q]^{2/q}) \leq C_a \mathcal{E}(g, g), \quad (5.2)$$

The exact open question on page 42 of the source.

The same paper supplies an especially useful test case. For $0 < \eta < 1/2$, let

$$K_1 = \begin{pmatrix} \sqrt{\eta} & 0 \\ 0 & \sqrt{1-\eta} \end{pmatrix}, \quad K_2 = \begin{pmatrix} 0 & \sqrt{\eta} \\ \sqrt{1-\eta} & 0 \end{pmatrix},$$

and $\Phi(X) = K_1^* X K_1 + K_2^* X K_2$. Its invariant state is $\sigma = \text{diag}(\eta, 1 - \eta)$, and $\Phi - \text{id}$ is primitive and $[\sigma]_{p,0}$ detailed balanced, but not GNS detailed balanced.

$$\partial_t \varphi(A_t, B_t)(\cdot) = (\delta_1 \varphi)((A_t, A_t), B_t)[\partial_t A_t, \cdot] + (\delta_2 \varphi)(A_t, (B_t, B_t))[\cdot, \partial_t B_t].$$

APPENDIX B. NOTE ON THE DETAILED BALANCE CONDITION

Noting (4.13), it follows from Lemma 2.2 that the QMS satisfying σ -GNS DBC is also self-adjoint with respect to the inner product $\langle \cdot, \cdot \rangle_{[\sigma]_{p,0}}$. In this appendix, we modify the discussions in [36, Appendix B] and [20, Appendix B] to show that the $[\sigma]_{p,0}$ -DBC, for $p \in (1, 2)$, and KMS DBC are not comparable, and that there exists a primitive QMS satisfying $[\sigma]_{p,0}$ -DBC but not σ -GNS DBC.

Let $\{|0\rangle, |1\rangle\}$ be the standard basis of $\mathbb{C}^2$, and $|v_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|v_2\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2|1\rangle)$ be another basis of $\mathbb{C}^2$. We define the quantum channel $\Phi(X) = K_1^* X K_1 + K_2^* X K_2$ with $K_1 = |v_1\rangle\langle 0|$ and $K_2 = |v_2\rangle\langle 1|$. It is easy to see that the associated unique invariant state of $\Phi^\dagger$ is

$$\sigma = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix},$$

and the spectrum of $\Phi^\dagger$ is given by $\{1, \frac{3}{10}, 0\}$ with 0 of multiplicity two. We denote by $\Phi_{\text{KMS}}^\dagger$ the adjoint of Φ with respect to the KMS inner product and define $\Psi = \Phi_{\text{KMS}}^\dagger \Phi$, which is also a quantum channel with $\Psi^\dagger(\sigma) = \sigma$ but satisfying the KMS DBC. Note that for a linear map Φ on $\mathcal{B}(\mathcal{H})$ satisfying both $[\sigma]_{p,0}$ -DBC and KMS DBC, there holds $g(\Delta_\sigma) \circ \Phi = \Phi \circ g(\Delta_\sigma)$, where $g(x) = \kappa_{1/p}(x)x^{1/2}$ and $g(\Delta_\sigma) = [\sigma]_{p,0}^{-1} \Gamma_\sigma$. By definition (A.4), we have $g(1/x) = g(x)$. Then, by exactly the same argument as in [20, Appendix B], we can show that the operator Ψ defined above does not commute with $g(\Delta_\sigma)$ and hence the Lindbladian $\Psi - \text{id}$ satisfies the KMS DBC but not $[\sigma]_{p,0}$ -DBC.

We next consider $\tilde{\Psi} := [\sigma]_{p,0}^{-1} \circ \Psi^\dagger \circ \Gamma_\sigma$, which is a quantum channel, since the operator $[\sigma]_{p,0}^{-1}$ is completely positive for $p \in (1, 2)$ by [68, Example 4.7]. Then we can check from definition that $\tilde{\Psi}$ satisfies the $[\sigma]_{p,0}$ -DBC. Again, since Ψ defined above does not commute with $g(\Delta_\sigma)$, neither does $\tilde{\Psi}$. Hence, the Lindbladian $\tilde{\Psi} - \text{id}$ satisfies $[\sigma]_{p,0}$ -DBC but not KMS DBC.

We conclude with an example of a primitive QMS with $[\sigma]_{p,0}$ -DBC but not σ -GNS DBC. The construction is modified from [20] as well. We define, for some $\eta \in (0, 1/2)$,

$$K_1 = \begin{bmatrix} \sqrt{\eta} & 0 \\ 0 & \sqrt{1-\eta} \end{bmatrix}, \quad K_2 = \begin{bmatrix} 0 & \sqrt{\eta} \\ \sqrt{1-\eta} & 0 \end{bmatrix},$$

and the associated quantum channel $\Phi(X) = K_1^* X K_1 + K_2^* X K_2$, which has the unique invariant state:

$$\sigma = \begin{bmatrix} \eta & 0 \\ 0 & 1-\eta \end{bmatrix}.$$

It is direct to verify $[\sigma]_{p,0}^{-1} \circ \Phi^\dagger = \Phi \circ [\sigma]_{p,0}^{-1}$, i.e., Φ satisfies the $[\sigma]_{p,0}$ -DBC. However, it was shown in [20] that σ -GNS DBC does not hold for Φ . It follows that the generator $\Phi - \text{id}$ is the desired example.

APPENDIX C. RECKNER'S INEQUALITY FOR NON-PRIMITIVE QMS

The source's non-GNS primitive example, page 44.

2 The partial theorem

For $1 < p < \infty$ and $q = p/(p-1)$, write

$$\|X\|_{p,\sigma} = \left(\text{Tr} \left| \Gamma_\sigma^{1/p}(X) \right|^p \right)^{1/p}, \quad I_{q,p}(X) = \Gamma_\sigma^{-1/q} \left(\Gamma_\sigma^{1/p}(X)^{p-1} \right)$$

for $X > 0$. The source's normalization is

$$\mathcal{E}_{p,\mathcal{L}}(X) = -\frac{pq}{4} \langle I_{q,p}(X), \mathcal{L}X \rangle_{\sigma,1/2}, \quad \langle A, B \rangle_{\sigma,1/2} = \text{Tr}(\sigma^{1/2} A^* \sigma^{1/2} B).$$

Theorem 1 (Poisson-type strict positivity). *Let $\Phi_1, \dots, \Phi_m : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ be unital completely positive maps satisfying $\Phi_k^\dagger(\sigma) = \sigma$, let $\gamma_k > 0$, and set*

$$\mathcal{L} = \sum_{k=1}^m \gamma_k (\Phi_k - \text{id}).$$

Then $\mathcal{E}_{p,\mathcal{L}}(X) \geq 0$ for every $X > 0$ and every $1 < p < \infty$. Equality holds precisely when $\Phi_k(X) = X$ for every k . Consequently, if $\mathcal{L}$ is primitive, then

$$\mathcal{E}_{p,\mathcal{L}}(\Gamma_\sigma^{-1}(\rho)) > 0 \quad \text{for every full-rank state } \rho \neq \sigma.$$

No detailed-balance assumption is needed for this conclusion.

Proof. First consider one map Φ . It is a contraction on every weighted $L_p(\sigma)$ space:

$$\|\Phi X\|_{p,\sigma} \leq \|X\|_{p,\sigma}. \quad (2)$$

For completeness, at $p = \infty$ this is the usual operator-norm contraction of a unital positive map. At $p = 1$, the conjugated map

$$\widehat{\Phi} = \Gamma_\sigma \circ \Phi \circ \Gamma_\sigma^{-1}$$

is completely positive and trace preserving: using $\Phi^\dagger(\sigma) = \sigma$ gives $\text{Tr } \widehat{\Phi}(A) = \text{Tr } A$. Hence it contracts the trace norm. Complex interpolation between these two endpoint spaces gives (2); the interpolated norm is exactly $\|X\|_{p,\sigma} = \|\Gamma_\sigma^{1/p} X\|_p$.

Put $Y = \Gamma_\sigma^{1/p}(X) > 0$. The weighted duality pairing satisfies

$$\begin{aligned} \langle I_{q,p}(X), X \rangle_{\sigma,1/2} &= \text{Tr } Y^p = \|X\|_{p,\sigma}^p, \\ \langle I_{q,p}(X), \Phi X \rangle_{\sigma,1/2} &= \text{Tr} \left(Y^{p-1} \Gamma_\sigma^{1/p}(\Phi X) \right). \end{aligned}$$

Schatten Hölder and (2) therefore yield

$$\begin{aligned} \langle I_{q,p}(X), \Phi X \rangle_{\sigma,1/2} &\leq \|Y^{p-1}\|_q \|\Gamma_\sigma^{1/p}(\Phi X)\|_p \\ &\leq \|X\|_{p,\sigma}^{p-1} \|X\|_{p,\sigma} = \langle I_{q,p}(X), X \rangle_{\sigma,1/2}. \end{aligned} \quad (3)$$

Thus $\mathcal{E}_{p,\Phi-\text{id}}(X) \geq 0$.

If equality holds, equality holds in the first line of (3). Since $1 < p < \infty$ and both matrices are positive, the equality case of Schatten Hölder gives $\Gamma_\sigma^{1/p}(\Phi X) = cY$ for some $c \geq 0$. Equality with the leftmost quantity in (3) forces $c = 1$, hence $\Phi X = X$. The converse is immediate.

The Dirichlet form is linear in the generator, so

$$\mathcal{E}_{p,\mathcal{L}}(X) = \sum_{k=1}^m \gamma_k \mathcal{E}_{p,\Phi_k-\text{id}}(X).$$

Every summand is nonnegative, and their sum vanishes exactly at their common fixed points. If $\mathcal{L}$ is primitive, its fixed-point algebra is $\mathbb{C}I$, so equality forces $X = cI$. Finally, when $X = \Gamma_\sigma^{-1}(\rho)$ and $\text{Tr } \rho = 1$, the normalization gives $c = 1$ and $\rho = \sigma$. $\square$

3 Application to the paper's missing example

The explicit Appendix B channel displayed above is unital, fixes σ in the dual, and $\Phi - \text{id}$ is primitive. Theorem 1 therefore gives (1) for this $[\sigma]_{p,0}$ -DBC but non-GNS family, for every $0 < \eta < 1/2$ and every $1 < p < \infty$. Thus the paper's example showing that the two detailed-balance classes differ does not create an exception to strict positivity.

The same argument applies to any generator that can be uniformized, meaning $\text{id} + \varepsilon \mathcal{L}$ is a unital completely positive map for some $\varepsilon > 0$: then $\mathcal{L} = \varepsilon^{-1}((\text{id} + \varepsilon \mathcal{L}) - \text{id})$.

4 Upgrade attempts and the remaining obstruction

The strongest extension route was tested rather than silently assumed.

1. The two non-GNS qubit constructions in Appendix B were implemented directly. Differential evolution over the full-rank Bloch ball, at 35 parameter choices, found the unique zero at the invariant state, up to 2×10^{-15} floating-point error.
2. A general qubit GKSL generator was parametrized by a Hermitian positive 3×3 Kossakowski matrix and a traceless Hamiltonian. Imposing $[\sigma]_{p,0}$ -DBC gives a six-dimensional real cone. For 20 sampled primitive generators at $(\eta, p) = (0.2, 1.5)$, global minimization again found only the invariant-state zero, at residuals below 8×10^{-17} .
3. Uniformization does not cover that whole cone. The primitive $[\sigma]_{p,0}$ -DBC boundary generator with Kossakowski matrix

$$C = \begin{pmatrix} 1/2 & -0.3i & 0 \\ 0.3i & 1/2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (\sigma = \text{diag}(0.2, 0.8), p = 3/2)$$

has spectrum $\{0, -1, -1/2, -1/2\}$, but the Choi matrix of $\text{id} + \varepsilon \mathcal{L}$ has a negative eigenvalue for every tested $\varepsilon > 0$, asymptotic to $-0.045\varepsilon^2$. Hence the proof above cannot be extended to all generators by taking one Euler step.

4. Passing instead to the genuine channel $e^{t\mathcal{L}}$ proves positivity of the finite difference $\langle I_{q,p}(X), (I - e^{t\mathcal{L}})X \rangle$, but after division by t and $t \downarrow 0$ strictness may be lost. A contraction can touch a smooth norm sphere to first order while entering it at second order. This is the exact infinitesimal equality obstruction left by the argument.

The computations are reproducible from the two scripts in `code/`. They are evidence and obstruction checks; they are not used in the proof of Theorem 1.

5 Bounded novelty check and conclusion

A bounded search through 11 August 2026 covered the run indexes, the exact open sentence, the paper title and arXiv identifier with the relevant detailed-balance and Dirichlet-form phrases, and the four later works indexed as citing the published article. None claimed this strict-positivity result or a resolution of the general question. Novelty confidence is moderate: the proof uses standard weighted L_p contractivity and may be known as folklore, but its application to the source’s non-GNS example does not appear to have been recorded.

The general question remains open, but Theorem 1 removes a broad and natural class, including the paper’s own separating example, and isolates why the same two-line argument does not automatically cover every Lindbladian.

References

- [1] B. Li and J. Lu, *Interpolation between modified logarithmic Sobolev and Poincaré inequalities for quantum Markovian dynamics*, J. Stat. Phys. 190 (2023), 161; arXiv:2207.06422. <https://arxiv.org/abs/2207.06422>.
- [2] S. Beigi, N. Datta, and C. Rouzé, *Quantum reverse hypercontractivity: its tensorization and application to strong converses*, Commun. Math. Phys. 376 (2020), 753–794; arXiv:1804.10100. <https://arxiv.org/abs/1804.10100>.